

# Search Engine Ranking Game Student Handout

TOK Cognitive Science Activity Bank • Knowledge and Technology • 25-30 min

|            |                          |      |                            |
|------------|--------------------------|------|----------------------------|
| Category   | Knowledge and Technology | Time | 25-30 min                  |
| Difficulty | Medium                   | ID   | search-engine-ranking-game |

Big idea: Visibility is not the same as reliability.

## Student-Facing Prompt

Rank the six search results from most reliable to least reliable. Then identify which ranking signal influenced you most: expertise, popularity, freshness, sponsorship, relevance, or emotional appeal.

## Concrete Stimulus Packet

|                                                                   |                                                            |                                                                                                                            |
|-------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Result 1: Ad · PureGreen Cleanse Co.</b>                       | "7-Day Celery Juice Liver Detox — feel lighter fast."      | Sells detox juice packs. No author listed. Ranking signal: Sponsorship.                                                    |
| <b>Result 2: WellnessNow Blog</b>                                 | "I drank celery juice every morning. Here's what changed." | Personal story, dramatic before/after claims, no medical sources. Ranking signal: Emotional appeal.                        |
| <b>Result 3: National Liver Foundation — Patient Guide</b>        | "Your liver already removes toxins from your blood."       | Explains liver function; reviewed by medical advisers. Ranking signal: Expertise.                                          |
| <b>Result 4: HealthLine Daily</b>                                 | "Celery juice: benefits, myths, and what evidence says."   | Notes hydration/nutrients but says detox claims are unsupported. Ranking signal: Relevance.                                |
| <b>Result 5: TikTok Clip: @CleanBodyGuru</b>                      | "Doctors don't want you to know this detox trick."         | 2.1M views, no evidence provided. Ranking signal: Popularity.                                                              |
| <b>Result 6: Research Summary: Journal of Nutrition Education</b> | "Vegetable intake and health outcomes."                    | Peer-reviewed; does not test celery juice detox claims directly. Ranking signal: Academic credibility, indirect relevance. |

## Actual Lesson Questions

| # | Question for students                                     |
|---|-----------------------------------------------------------|
| 1 | Which result would you click first, and why?              |
| 2 | Which result is most reliable for the query?              |
| 3 | Which result might a popularity-based ranking put first?  |
| 4 | Which result might a sponsorship-based ranking put first? |
| 5 | Which result best shows expertise?                        |

| # | Question for students                                 |
|---|-------------------------------------------------------|
| 6 | What does this show about visibility and reliability? |

## Ready-to-Use Examples

- Mock search results ranked by popularity, expertise, freshness, sponsorship, or relevance.
- A query about a health claim with credible, vague, commercial, and opinion sources.
- Two ranking rules that produce different 'top answers' for the same searcher.

## Student Task

Query: Does drinking celery juice detox your liver? Rank all six results from most reliable to least reliable, then explain which signal shaped your ranking.

## Activity Steps

1. Present a question such as "Is homework effective?"
2. Students rank which results they would click first.
3. Reveal hidden information: sponsored result, old source, expert review, blog post, advocacy group and research article.
4. Students redesign the search page to make reliability easier to judge.

## Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

| What I noticed | What I inferred | Evidence | Confidence |
|----------------|-----------------|----------|------------|
|                |                 |          |            |
|                |                 |          |            |
|                |                 |          |            |

## TOK Debrief

- Knowledge question: Does visibility affect what people accept as true?
- Is the first answer usually the best answer?
- How should search engines balance relevance, profit and public knowledge?

## Knowledge Questions

- Does visibility affect what people accept as true?
- Is the first answer usually the best answer?
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## Transfer Example

For example, run this activity with mock search results ranked by popularity, expertise, freshness, sponsorship, or relevance. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

## Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...